

Forearm

Department of anatomy

Musculoskeletal System

Second Part

The Upper Limb (6)

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Lecture points

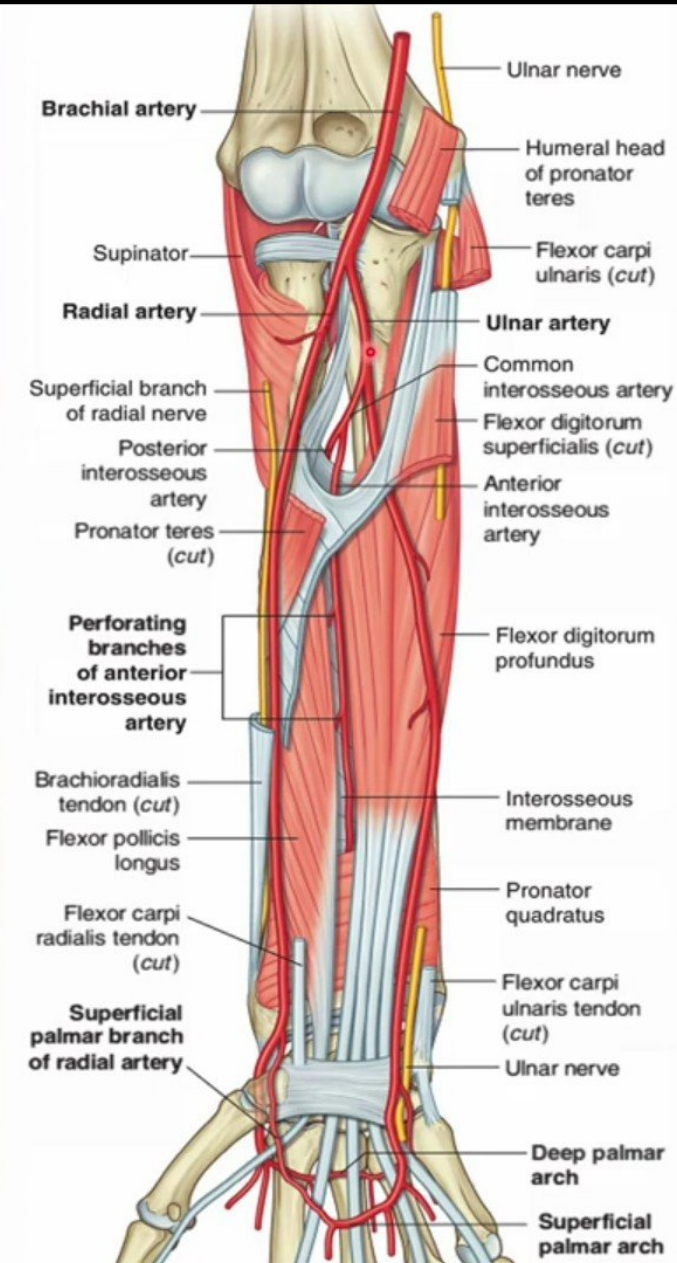
- ✓ Introduction
- ✓ Compartments.
- Muscles of the Forearm
- ✓ Contents of the Anterior Fascial Compartment of the Forearm
 - ✓ Muscles.
- ✓ Contents of the Posterior Fascial Compartment of the Forearm.
 - ✓ Muscles.
- Blood supply in the forearm:
 - Ulnar artery in the forearm.
 - Anterior interosseous artery.
 - Posterior interosseous artery.
 - Radial artery in the forearm.
 - Surface Marking of ulnar and radial artery.
- ✓ Summary of arteries of the Anterior and Posterior Compartment of the Forearm
- ✓ Venous drainage in the arm and Forearm
- ✓ Lymphatic drainage in the forearm
- Innervation in the Forearm.
- ✓ Median nerve in the anterior fascial compartment of the forearm.
- ✓ Anterior Interosseous Nerve .
- ✓ Ulnar nerve in the anterior fascial compartment of the forearm.
- ✓ Radial Nerve and the posterior interosseous nerve of the Forearm
- ✓ Innervation :
 - ✓ Sensory (dermatome).
 - ✓ Motor (myotome).

Blood supply in the forearm:
Ulnar artery
Radial artery
venous and lymphatic drainage

Ulnar artery in the forearm

Ulnar Artery

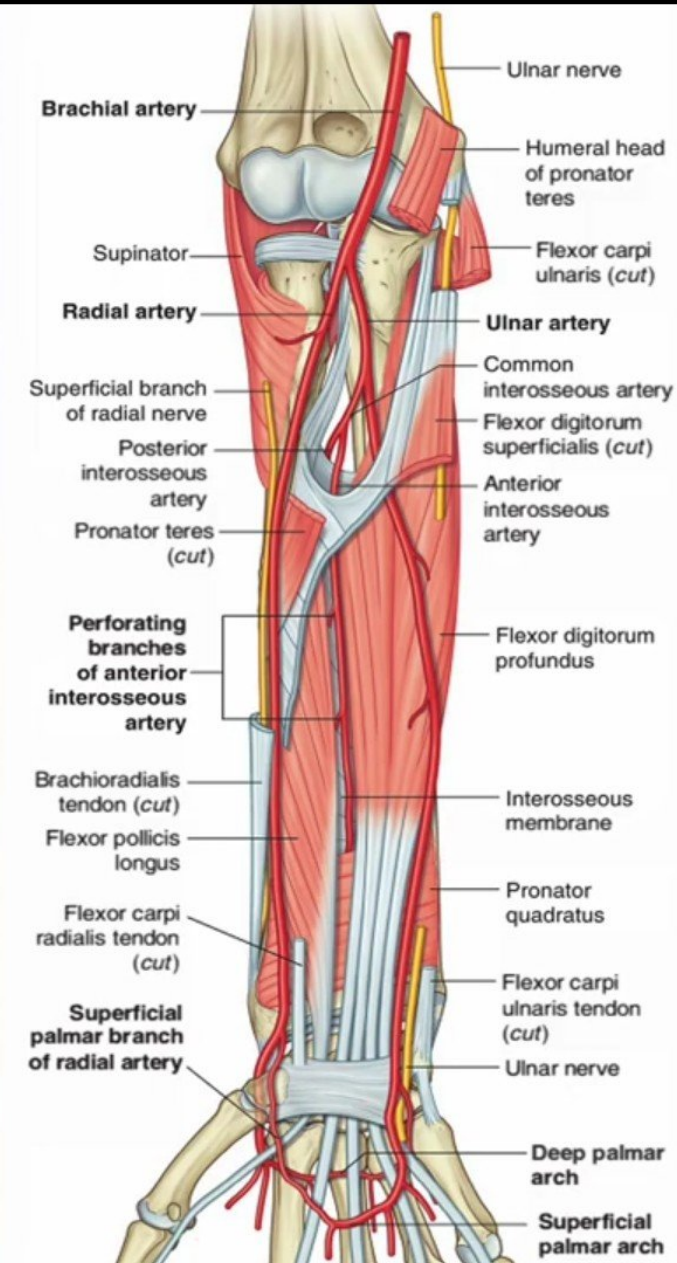
- It is **the larger** of the two terminal branches of **the brachial artery**.
- It begins in **the cubital fossa** at the **level of the neck of the radius**.
- It **descends through the anterior compartment** of the forearm and enters **the palm above the flexor retinaculum** in **company with** the **ulnar nerve**.
- It ends by forming the **superficial palmar arch**, often anastomosing with **the superficial palmar branch** of the **radial artery**.



Ulnar artery in the forearm

Ulnar Artery

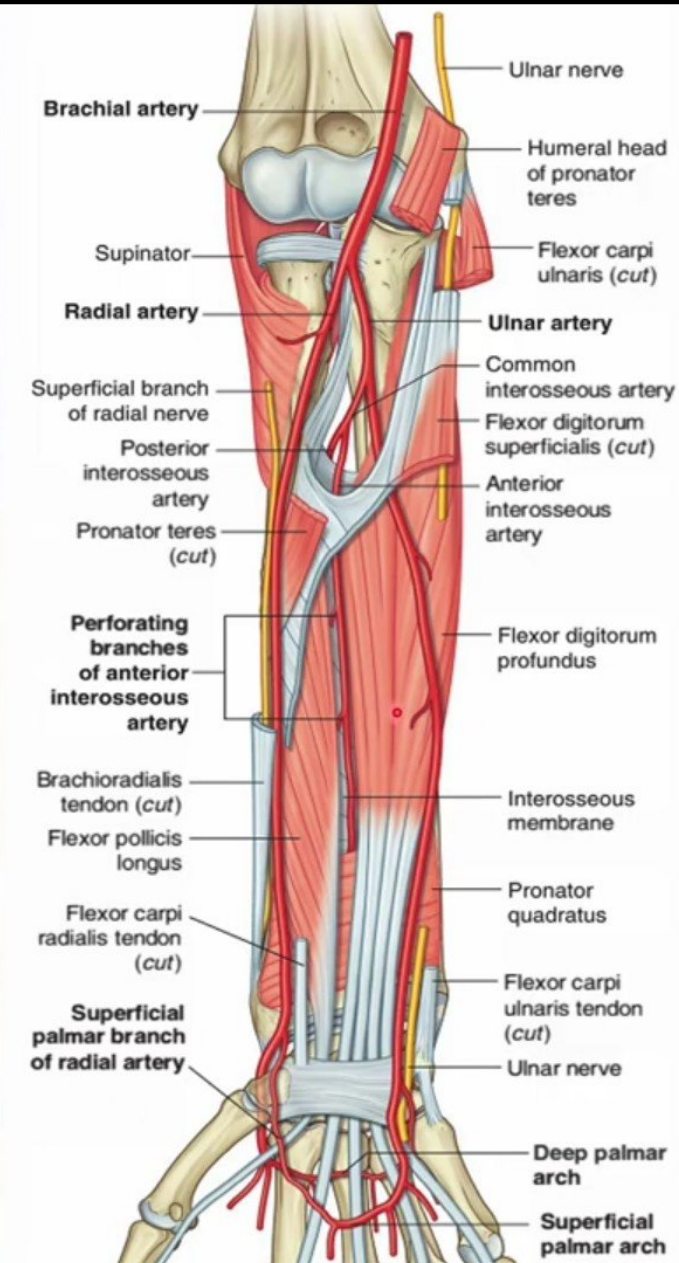
- In the **upper part of its course**, lies **deep to most of the flexor muscles**.
- **Below**, it becomes **superficial** and **lies between the tendons of the flexor carpi ulnaris and the tendons of the flexor digitorum superficialis**.
- **Above the flexor retinaculum**, it lies just **lateral to the pisiform bone** and is covered only by skin and fascia (site for **taking ulnar pulse**).



Ulnar artery in the forearm

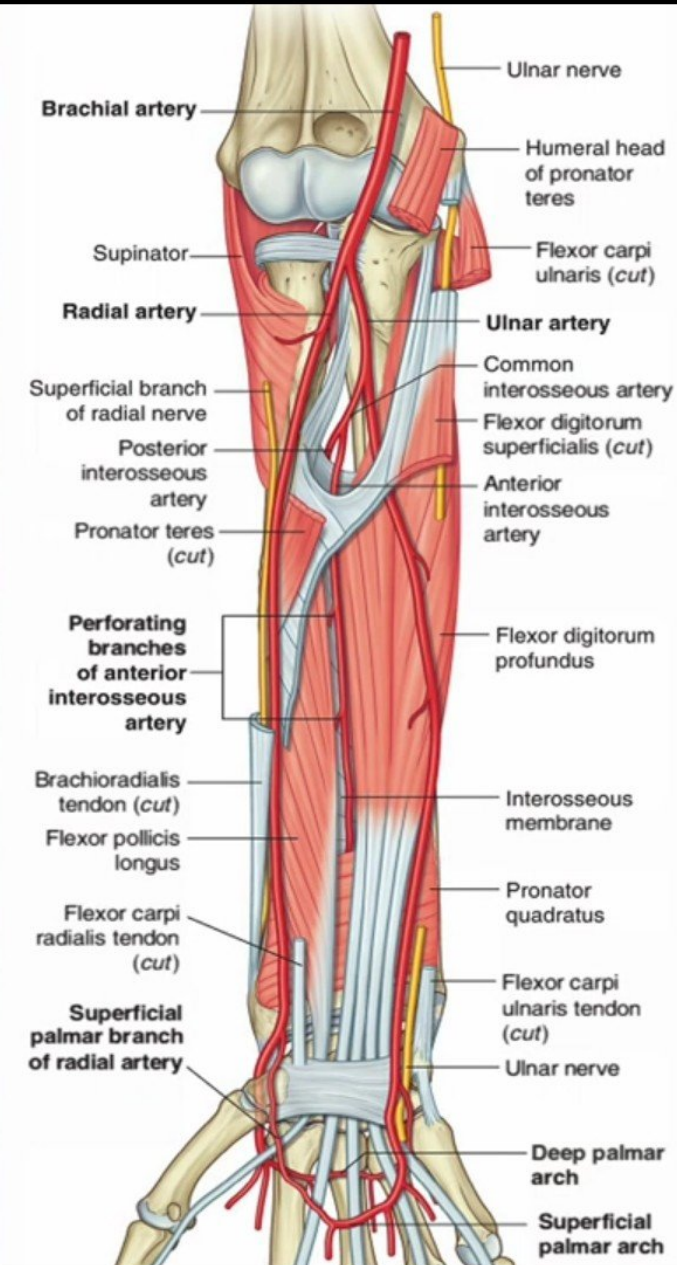
Branches of the ulnar artery:

- Muscular branches to neighboring muscles
- Recurrent branches that take part in the arterial anastomosis around the elbow joint.
- Branches that take part in the arterial anastomosis around the wrist joint.
- The common interosseous artery:
 - It **arises** from **the upper part of the ulnar artery** .
 - Its course divides into the **anterior** and **posterior interosseous arteries**.
 - The **interosseous arteries** are distributed to the muscles lying in front of and behind the interosseous membrane; they provide nutrient arteries to **the radius** and **ulna bone**.



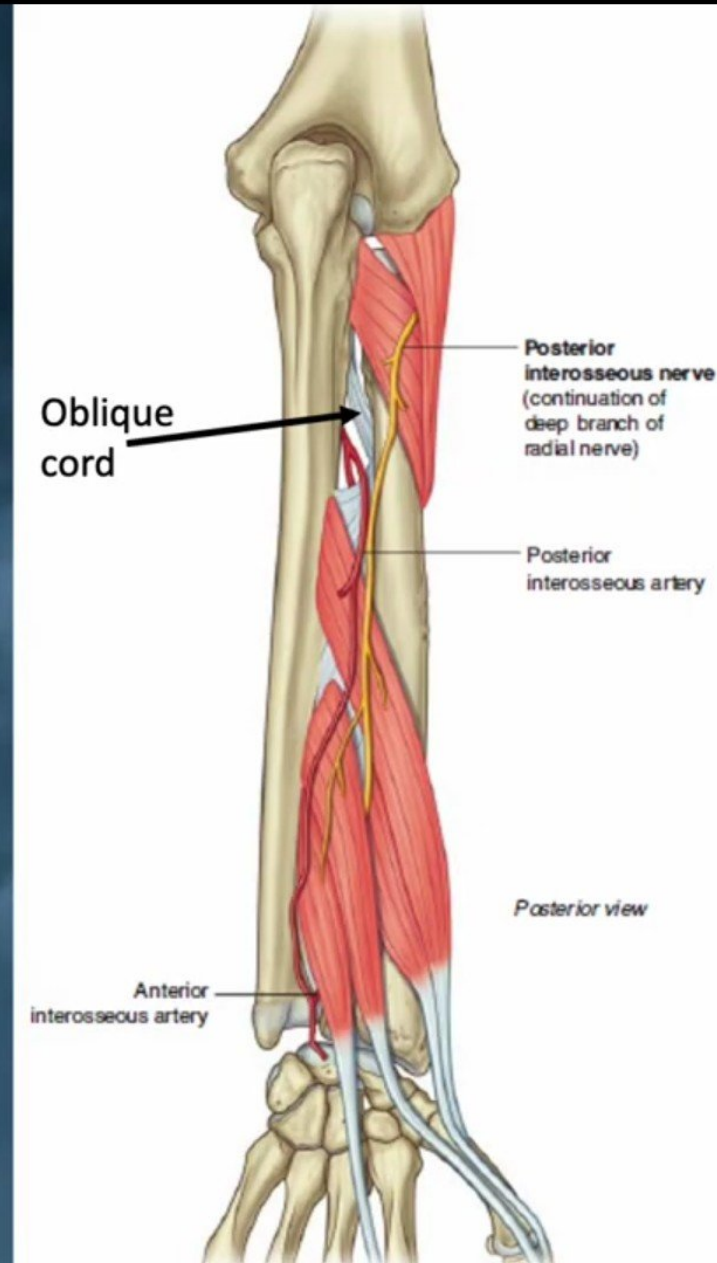
Anterior interosseous artery

- The **anterior interosseous artery** lies **deeply** on the **interosseous membrane** between **flexor digitorum profundus** and **flexor pollicis longus**, supplying each.
- **Perforating branches** pierce the **interosseous membrane** to supply the **deep extensor muscles**.
- **Nutrient vessels** are given to **both radius** and **ulna**.
- The artery passes posteriorly through **the interosseous membrane** at the level of the **upper border of pronator quadratus**.



Posterior interosseous artery

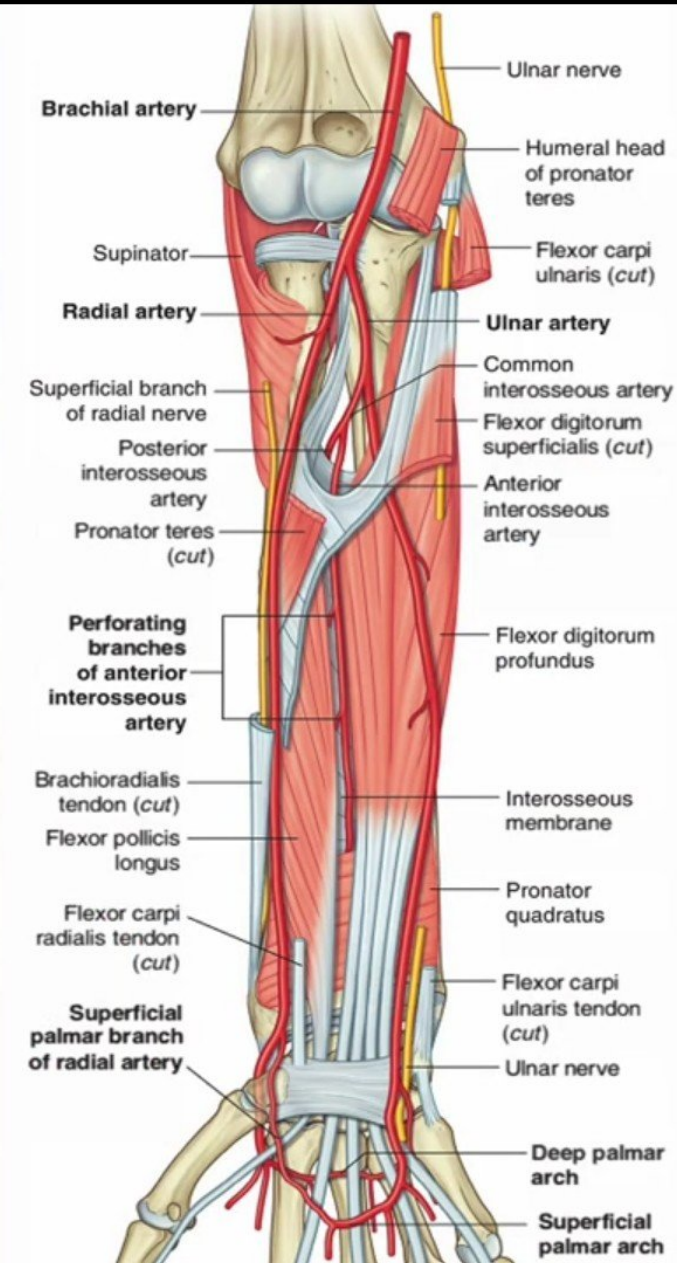
- The **posterior interosseous artery** disappears by passing backwards through the interosseous space between **the upper end of the interosseous membrane** and **the oblique cord** than enter the posterior compartment of forearm.
- It appears between the contiguous borders of **supinator muscle** and **the abductor pollicis longus muscle**, and runs down the back of the forearm between **the superficial and deep layers of muscles**, to both of which it distributes branches.
- It lies on **abductor pollicis longus muscle** and the **extensor pollicis brevis muscle**,
- It is accompanied by the **dorsal interosseous nerve**.
- At the lower part of the forearm it anastomoses with the dorsal carpal network.



Radial artery in the forearm

Radial Artery

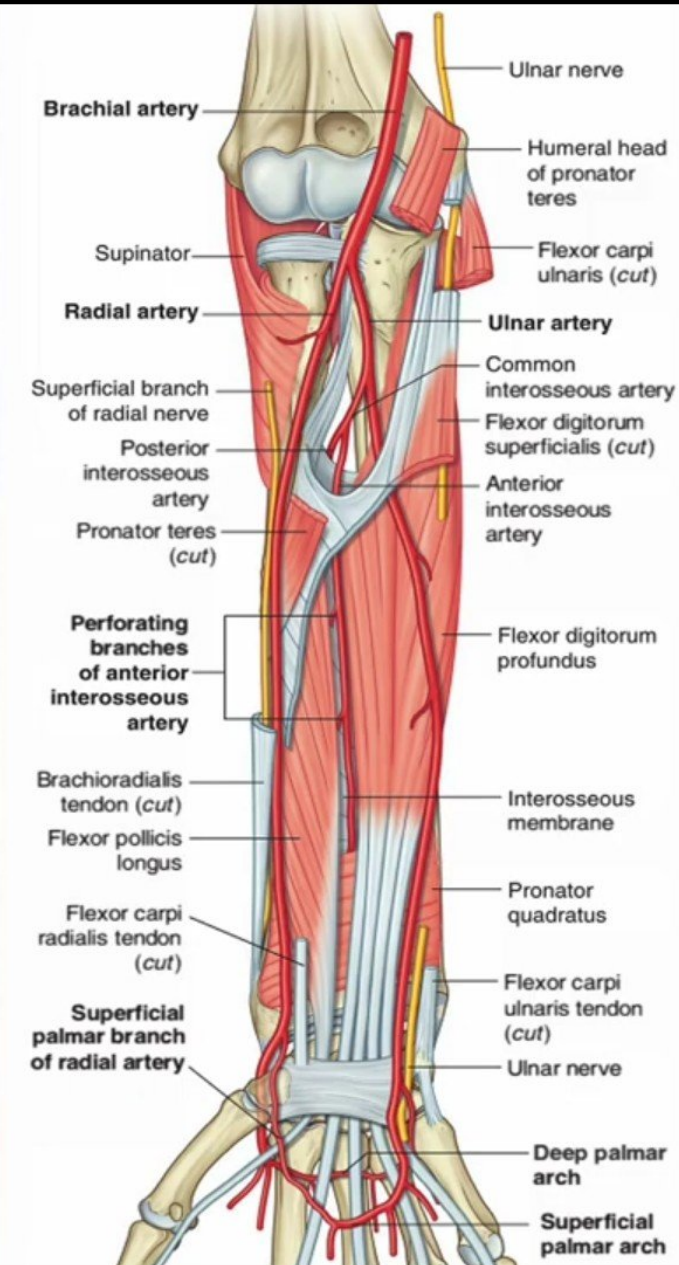
- It is **the smaller** of the terminal branches of **the brachial artery**.
- It **begins** in the **cubital fossa** at the **level of the neck of the radius**.
- It passes downward and **laterally, beneath the brachioradialis muscle** and resting on the deep muscles of the forearm.
- In **the middle third of its course**, the **superficial branch of the radial nerve** lies on its **lateral side**.



Radial artery in the forearm

In the distal part of the forearm:

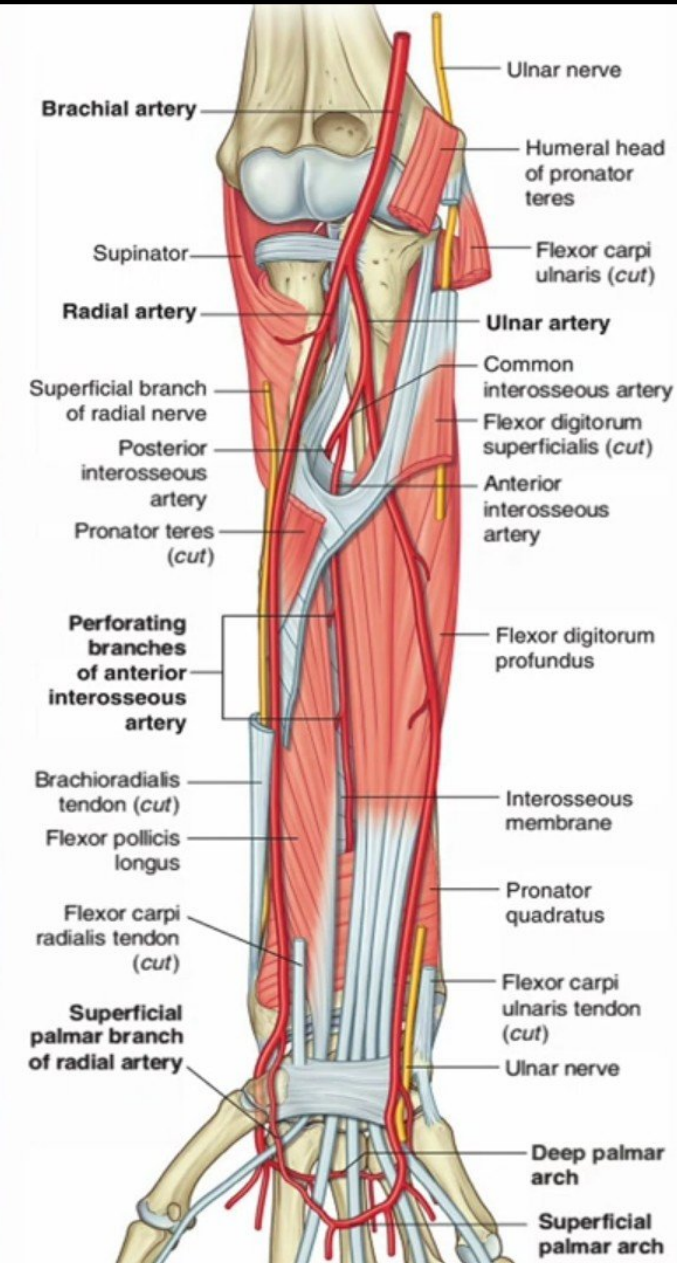
- The **radial artery** lies on the anterior surface of the radius and is covered only by skin and fascia.
- The **tendon of brachioradialis** found on lateral side of the radial artery and the **tendon of flexor carpi radialis** found on medial side of the radial artery (site for taking the **radial pulse**).
- The radial artery **leaves the forearm by winding around the lateral aspect of the wrist to reach the posterior surface of the hand.**



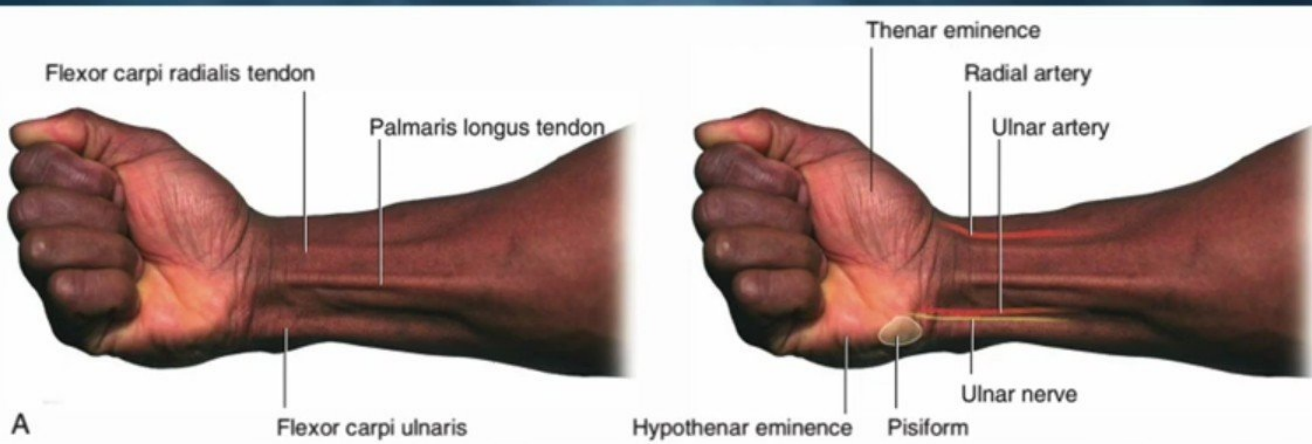
Radial artery in the forearm

Radial Artery branches in the Forearm

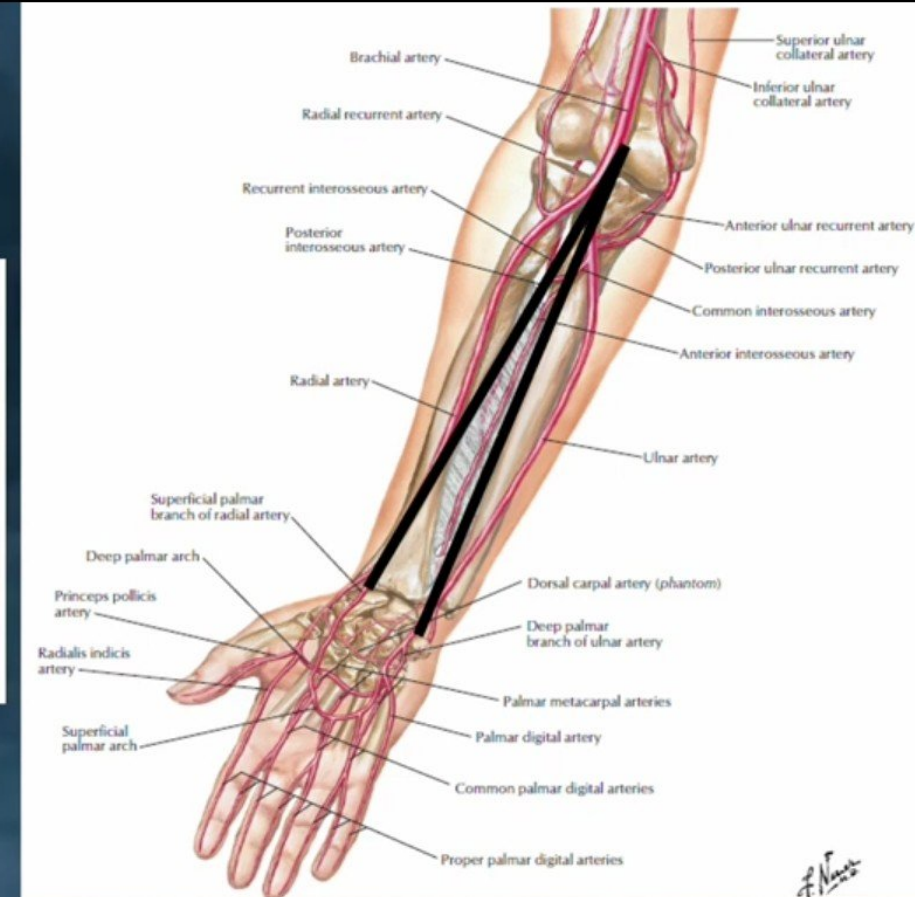
- **Muscular branches** to neighboring muscles.
- **Recurrent branch** which takes part in the arterial anastomosis around the elbow joint.
- **Small palmar carpal branch**, which contributes to anastomotic network of vessels that supply the carpal bones and joints.
- **Superficial palmar branch** which arises just above the wrist, enters the palm of the hand, and frequently joins the ulnar artery to form the **superficial palmar arch**.



Surface Marking



- The surface marking of **the radial artery** is along a line, slightly **convex laterally**, from medial to the biceps tendon in the cubital fossa to medial to the styloid process of the radius.
- It can be surgically exposed at its lower end which is the most common site for arterial cannulation.



- The *surface marking* of the **ulnar artery** is along a line, slightly **convex medially**, from medial to the biceps tendon in the cubital fossa to the radial side of the pisiform.
- It can be *surgically exposed* at the lower end and followed upwards by displacing flexor carpi ulnaris.
- The ulnar nerve on the ulnar side of the artery must be safeguarded.

Summary of Artery in the forearm

Radial artery

Origin:

In cubital fossa, as smaller terminal branch of brachial artery

Course/Distribution:

Runs distally under brachioradialis, lateral to flexor carpi radialis, defining boundary between the flexor and extensor compartments and supplying the radial aspect of both. Gives rise to a superficial palmar branch near the radiocarpal joint; it then transverse the anatomical snuff box to pass between the heads of the 1st dorsal interosseous muscle joining the deep branch of the ulnar artery to form the deep palmar arch

Ulnar artery

Origin: In cubital fossa, as larger terminal branch of brachial artery

Course/Distribution: Passes distally between 2nd and 3rd layers of forearm flexor muscles, supplying ulnar aspect of flexor compartment; passes superficial to flexor retinaculum at wrist, continuing as the superficial palmar arch (with superficial branch of radial) after its deep palmar branch joins the deep palmar arch

Radial recurrent artery

Origin: In cubital fossa, as 1st (lateral) branch of radial artery

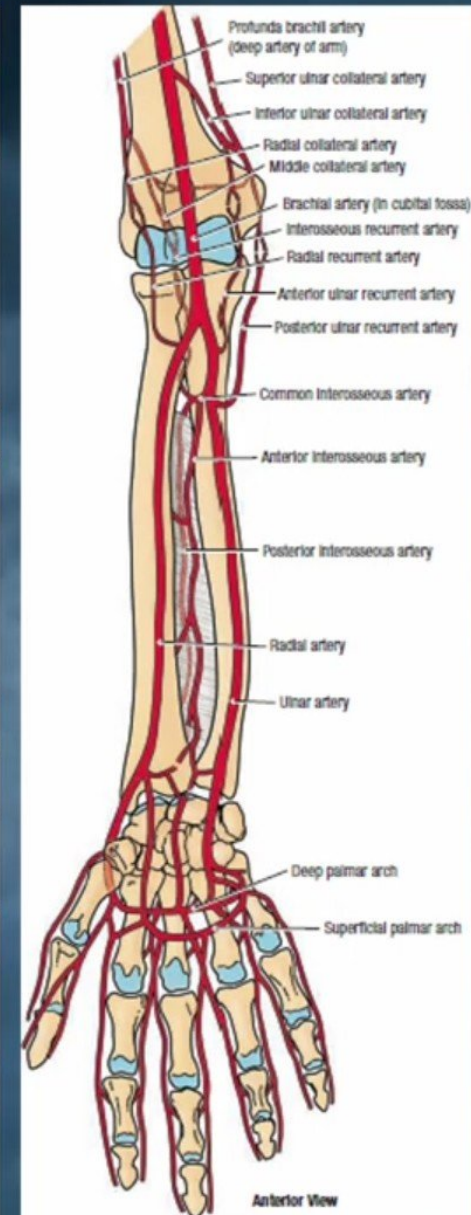
Course/Distribution: Courses proximally, superficial to supinator, passing between brachioradialis and brachialis to anastomose with radial collateral artery

Anterior and posterior ulnar recurrent arteries

Origin: In and immediately distal to cubital fossa, as 1st and 2nd medial branches of ulnar artery

Course/Distribution:

Course proximally to anastomose with the inferior and superior ulnar collateral arteries, respectively, forming collateral pathways anterior and posterior to the medial epicondyle of the humerus



Summary of artery in the forearm

Common interosseous artery

Origin: Immediately distal to the cubital fossa, as 1st lateral branch of ulnar artery

Course/Distribution: Terminates almost immediately, dividing into anterior and posterior interosseous arteries

Anterior and posterior interosseous arteries

Origin: Distal to radial tubercle, as terminal branches of common interosseous

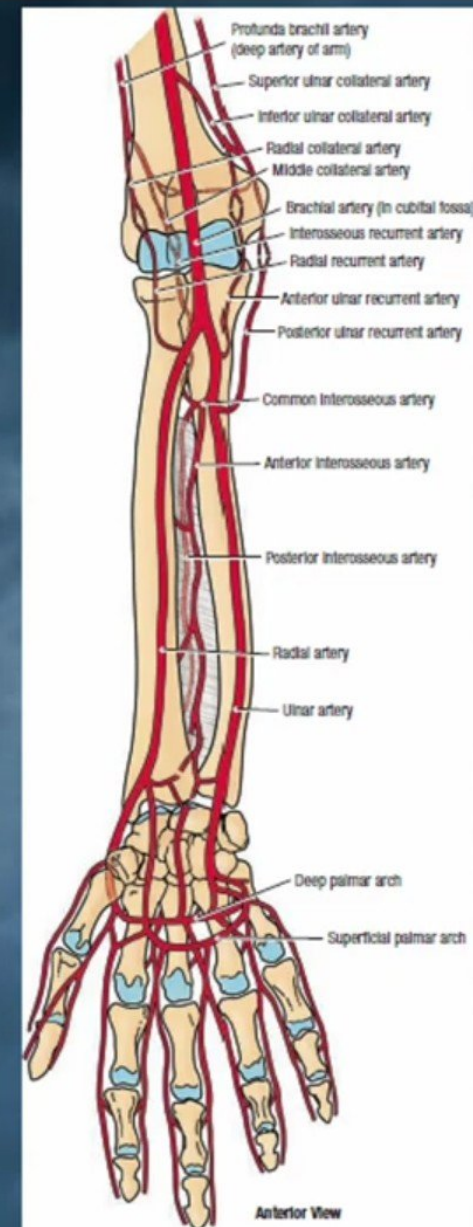
Course/Distribution: Pass to opposite sides of interosseous membrane; anterior artery runs on interosseous membrane; posterior artery runs between superficial and deep layers of extensor muscles as primary artery of compartment

Interosseous recurrent artery

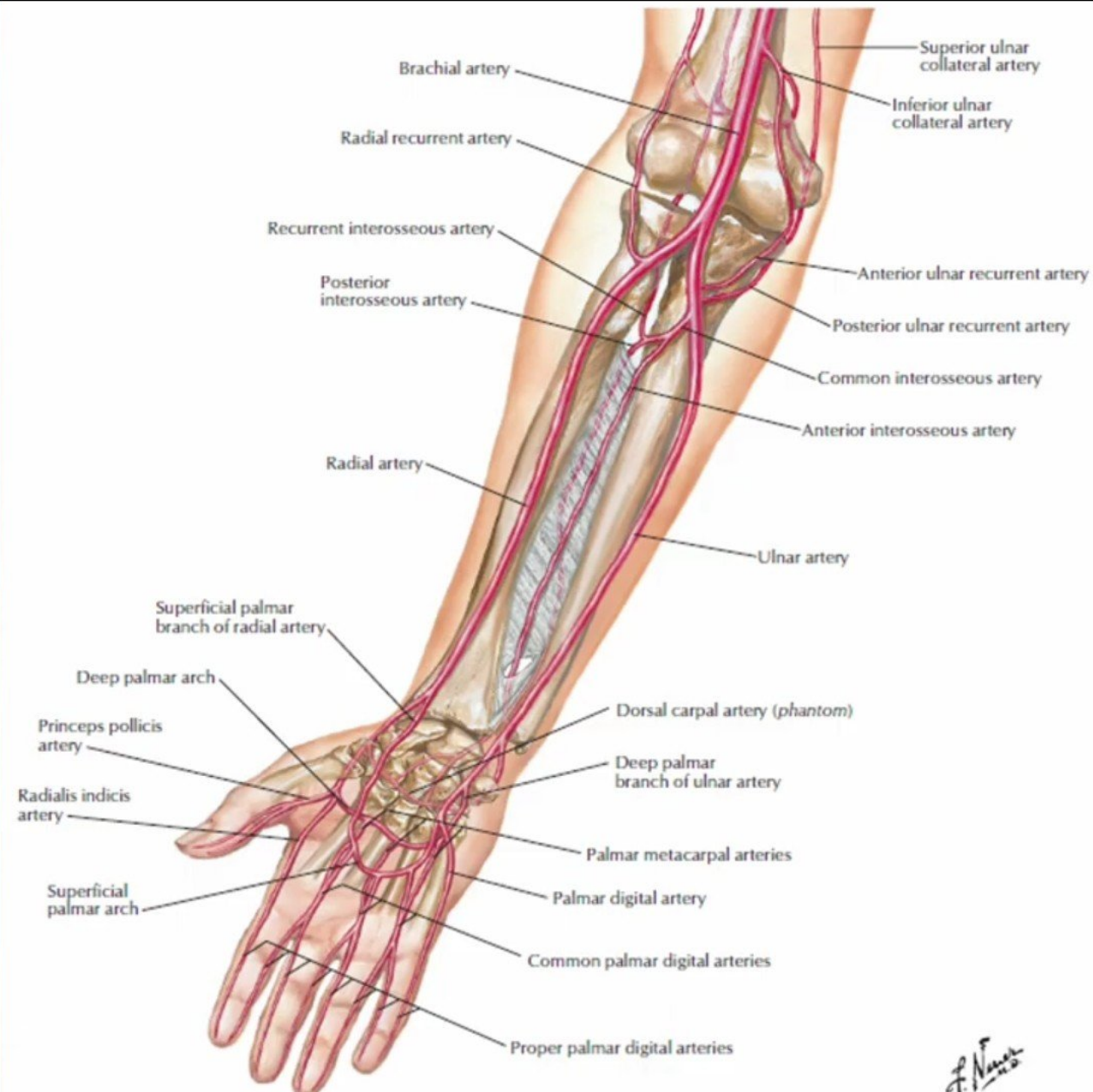
Origin: Initial part of posterior interosseous artery

Course/Distribution:

Courses proximally between lateral epicondyle and olecranon, deep to anconeus, to anastomose with middle collateral artery



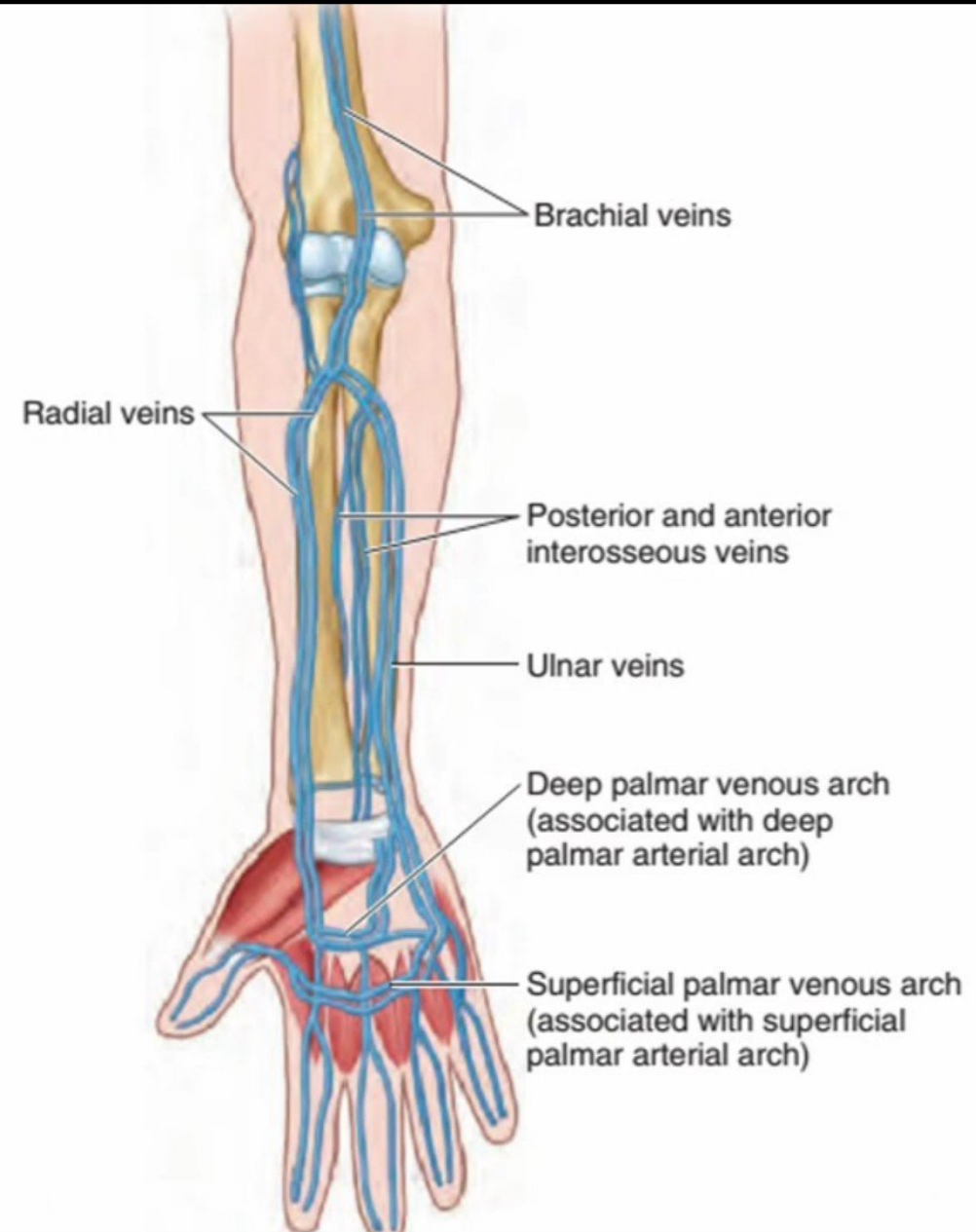
Summary of arteries of the Anterior and Posterior Compartment of the Forearm

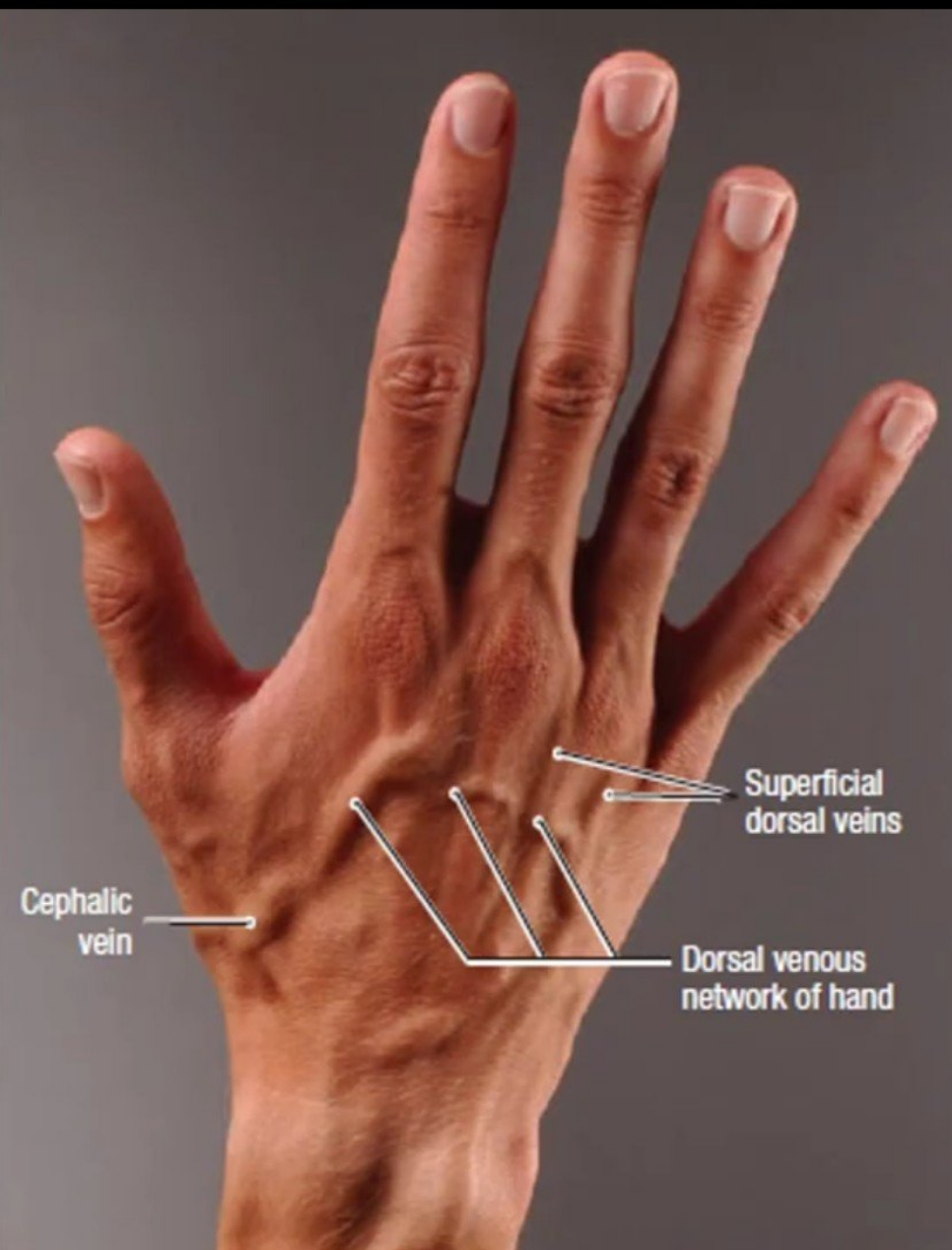
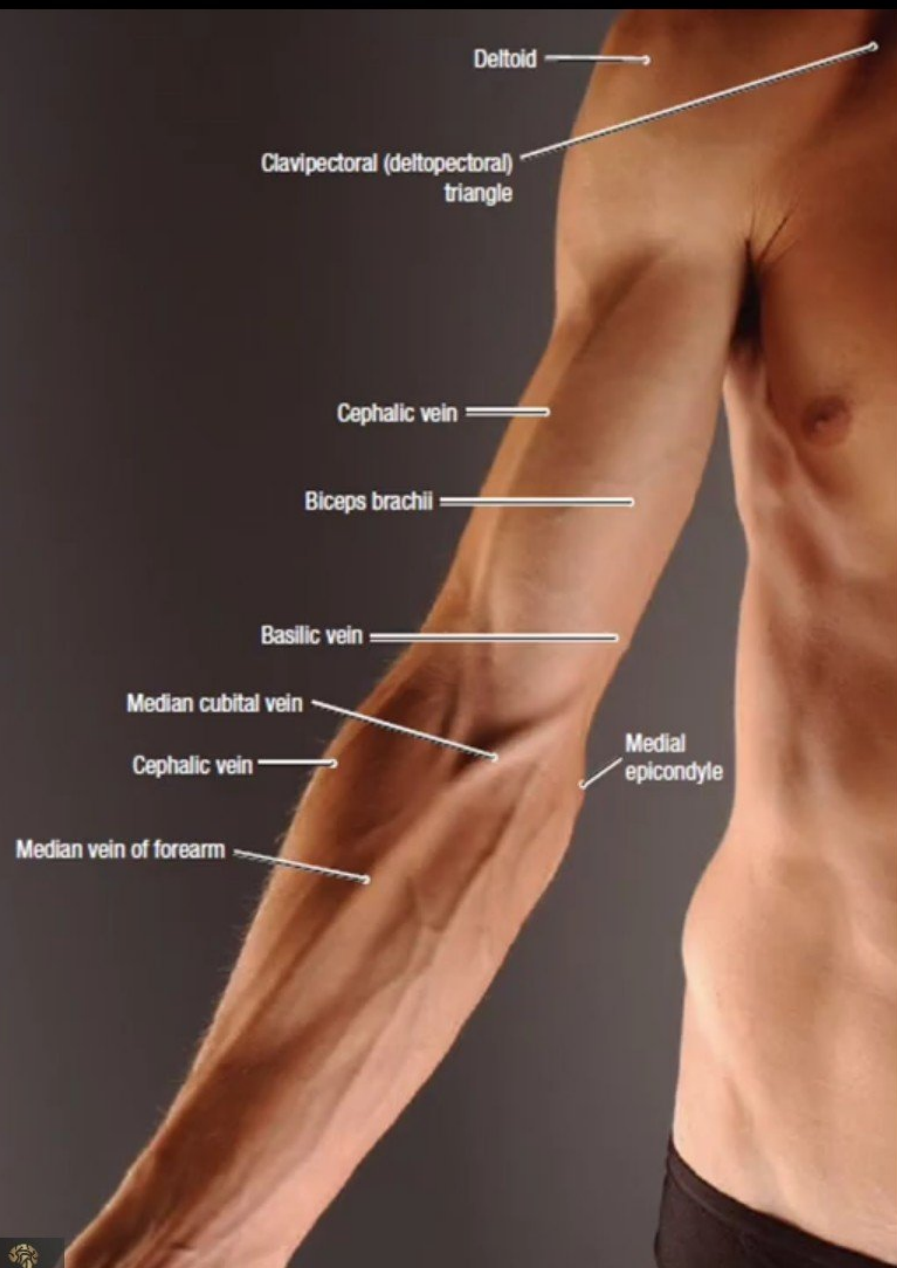


F. Nasser

Venous drainage in the arm and the Forearm

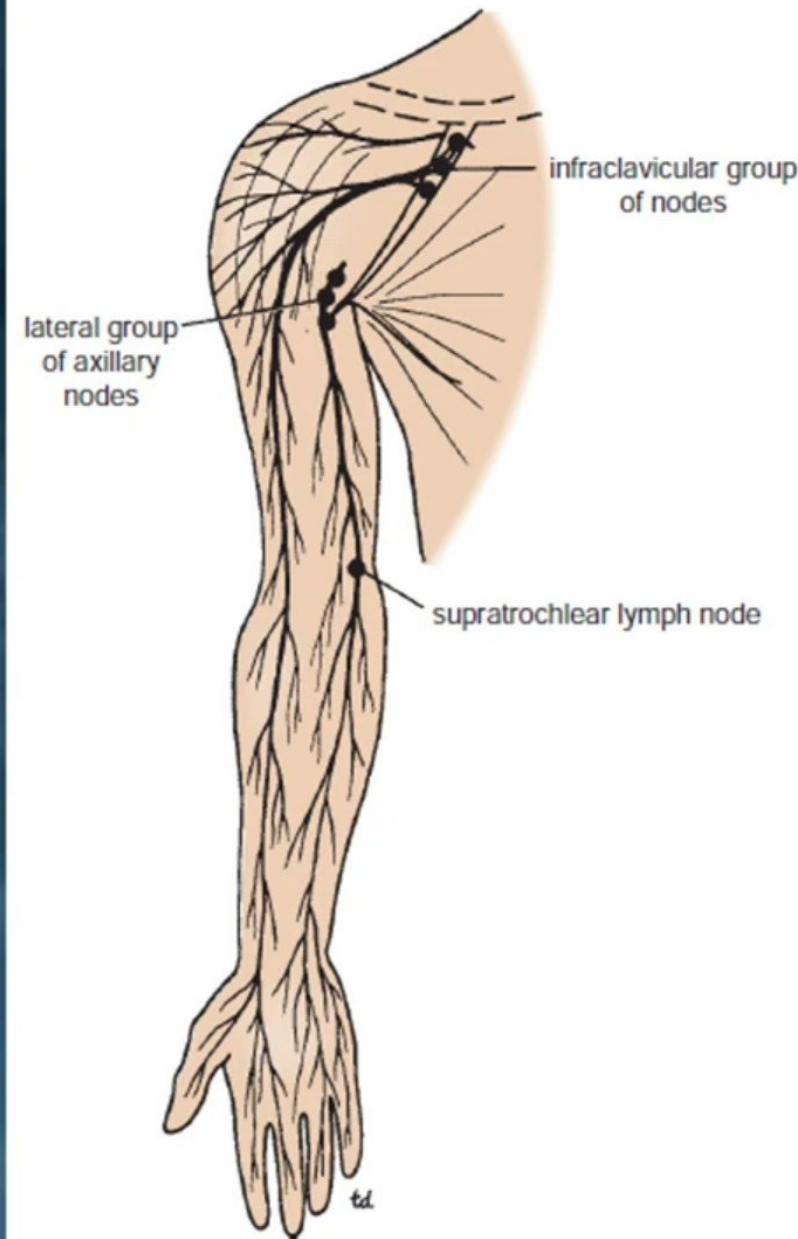
- Superficial(cephalic & basilic)
- Deep (two small veins accompany to arteries)





Lymphatic drainage in the forearm

- As elsewhere in the body:
 - ✓ The **superficial lymphatic's** follow **veins**.
 - ✓ The **deep lymphatic's** follow **arteries**.
- From **the ulnar side of the hand and forearm** the subcutaneous lymphatics run alongside the **basilica vein** to **the supratrochlear nodes**.
- From **the radial side** the lymphatics run alongside **the cephalic vein** to **the infraclavicular nodes**.
- From the **deep parts of the hand and forearm** and from **the supratrochlear nodes** lymphatics pass to the **lateral group of axillary nodes**.



*Will be continue ...
&
good luck..☺*